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Two-regular subgraphs of odd-uniform hypergraphs ☆

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ABSTRACT

Let $k \geq 3$ be an odd integer and let n be a sufficiently large integer. We prove that the maximum number of edges in an n-vertex k-uniform hypergraph containing no 2-regular subgraphs is $\binom{n-1}{k-1} + \lfloor \frac{n-1}{k} \rfloor$, and the equality holds if and only if H is a full k-star with center v together with a maximal matching omitting v. This verifies a conjecture of Mubayi and Verstraëte.

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1. Introduction

Turán problems are central in extremal graph theory. In general, Turán-type problems ask for the maximum number of edges of a (hyper)graph that does not contain certain subgraph(s). Their generalizations to hypergraphs appear to be extremely hard – for example, despite many existing works, the Turán density of a tetrahedron (four triples on four vertices) is still unknown (see [8]).

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Erdős [3] asked to determine the maximum size $f_k(n)$ of an n-vertex k-uniform hypergraph without any generalized 4-cycles, i.e., four distinct edges A, B, C, D such that $A \cup B = C \cup D$ and $A \cap B = C \cap D = \emptyset$. For $k = 2$, this reduces to a well-known problem of finding the Turán number for the 4-cycle. It is known that $f_2(n) = (1 + o(1))n^{3/2}$ [2,4] and the exact value of $f_2(n)$ for infinitely many n is obtained in [6]. For $k \geq 3$, Füredi [7] showed that $\binom{n-1}{k-1} + \lfloor \frac{n-1}{k} \rfloor \leq f_k(n) \leq \frac{7}{2}\binom{n}{k-1}$ [1] and conjectured the following.

Conjecture 1.1. *For $k \geq 4$ and $n \in \mathbb{N}$, $f_k(n) = \binom{n-1}{k-1} + \lfloor \frac{n-1}{k} \rfloor$.*

The lower bound is achieved by a full k-star together with a maximal matching omitting its center. Here a *full k-star* is a k-uniform n-vertex hypergraph which consists of all $\binom{n-1}{k-1}$ sets of size k containing a given vertex v, and the given vertex v is called the *center* of the full k-star. The most recent result on $f_k(n)$ is due to Pikhurko and Verstraëte [13], who showed that $f_k(n) \leq \min\{1 + 2/\sqrt{k}, 7/4\}\binom{n}{k-1}$, and $f_3(n) \leq \frac{13}{9}\binom{n}{2}$. This improves a result by Mubayi and Verstraëte [11]. In [9], the second author made a related conjecture about k-uniform hypergraphs containing no r pairs of disjoint sets with the same union when k is sufficiently bigger than r.

Since the generalized 4-cycles are 2-regular, i.e., each vertex has degree 2, one way to relax the original problem of Erdős is to consider the maximum size of n-vertex (hyper)graphs without any 2-regular sub(hyper)graphs (or more generally, without any r-regular subgraphs). In fact, the (relaxed) problem is of its own interest even for graphs. Pyber [14] proved that the largest number of edges in a graph with no r-regular subgraphs is $O(n \log n)$ for any $r \geq 2$, and in [15], Pyber, Rödl and Szemerédi showed that there are graphs with no r-regular subgraphs having $\Omega(n \log \log n)$ edges for any $r \geq 3$.

For non-uniform hypergraphs, it is easy to see that any hypergraph with no r-regular subgraphs has at most $2^{n-1} + r - 2$ edges and Kostochka and the second author [10] showed that if $n \geq \max\{425, r + 1\}$ then any n-vertex hypergraph with no r-regular subgraphs having the maximum number of edges must contain a vertex of degree 2^{n-1}. For uniform hypergraphs, the problem becomes more interesting. One natural candidate for the extremal example of k-uniform hypergraphs with no 2-regular subgraphs is the full k-star. Indeed, Mubayi and Verstraëte [12] proved the following.

Theorem 1.2. *([12]) For every even integer $k \geq 4$, there exists n_k such that the following holds for all $n \geq n_k$. If H is an n-vertex k-uniform hypergraph with no 2-regular subgraphs, then $|H| \leq \binom{n-1}{k-1}$. Moreover, equality holds if and only if H is a full k-star.*

In [9] the second author generalized the arguments in [12] and showed similar results for k-uniform hypergraphs with no r-regular subgraphs when $r \in \{3, 4\}$. Moreover, for

[1] In fact, Füredi [7] found a slightly better lower bound for $k = 3$, namely, $f_3(n) \geq \binom{n}{2}$ for $n \equiv 1$ or 5 mod 20.

odd k, Mubayi and Verstraëte [12] conjectured that $|H| \leq \binom{n-1}{k-1} + \lfloor \frac{n-1}{k} \rfloor$, and the only extremal graph is the full k-star plus a matching omitting the center of the star. In this paper, we prove this conjecture.

Theorem 1.3. *For every odd integer $k \geq 3$, there exists n_k such that the following holds for all $n \geq n_k$. If H is an n-vertex k-uniform hypergraph with no 2-regular subgraphs, then*

$$|H| \leq \binom{n-1}{k-1} + \left\lfloor \frac{n-1}{k} \right\rfloor.$$

Moreover, equality holds if and only if H is a full k-star with center v together with a maximal matching omitting v.

Theorem 1.2 [12] is proved via the stability approach introduced by Erdős and Simonovits [16], which has been widely used in extremal set theory. To prove Theorem 1.3, we also use the stability approach as well as some other ideas from [12]. One advantage when k is even is that there exist 2-regular k-uniform hypergraphs on $3k/2$ vertices. In contrast, for odd k, the smallest 2-regular k-uniform hypergraphs have order $2k$ and thus the analysis is more difficult (this is also the reason why more edges are allowed in the extremal graph for odd k, which makes the structure of the extremal examples more complicated). In our proof, we use some new tricks to prove the result with the absence of 2-regular k-uniform hypergraphs on $3k/2$ vertices.

2. Preliminaries

For a positive integer N we write $[N]$ to denote the set $\{1, \ldots, N\}$. We write $V(H)$ for the set of vertices, $E(H)$ for the set of edges of a hypergraph H. For a hypergraph H, we view H as a collection of edges, thus sometimes H refers to $E(H)$. We say that H is a *k-uniform hypergraph* or *k-uniform family* if every edge of H has size exactly k. Moreover, we always say *subgraph* instead of subhypergraph. For a hypergraph H and a set $S \subseteq V(H)$,

$$N_H(S) := \{e \setminus S : e \in E(H), S \subseteq e\} \quad \text{and} \quad d_H(S) = |N_H(S)|.$$

We say that a set S is an *s-set* if $|S| = s$. For a vertex $x \in V(H)$, we write $N_H(x) := N_H(\{x\})$ and $d_H(x) := d_H(\{x\})$. We say that $\{S, S'\}$ is an *equipartition* of a set A if $|S| = |S'|$, $S \cap S' = \emptyset$ and $S \cup S' = A$.

In order to prove Theorem 1.3, we use the following two theorems proved in [12]. These theorems give a rough structure of near-extremal hypergraphs.

Theorem 2.1. *([12]) For given $\varepsilon > 0$ and $k \in \mathbb{N}$, there exists $n_0 = n_0(k, \varepsilon)$ such that the following holds for all $n \geq n_0$. If H is an n-vertex k-uniform hypergraph with no 2-regular subgraphs, then*

$$|H| \le (1+\varepsilon)\binom{n-1}{k-1}.$$

Theorem 2.2. *([12]) For given $\varepsilon > 0$ and $k \in \mathbb{N}$, there exists $n_1 = n_1(k,\varepsilon)$ such that the following holds for all $n \ge n_1$. If H is an n-vertex k-uniform hypergraph with no 2-regular subgraphs with $|H| \ge \binom{n-1}{k-1}$, then H contains a vertex v with $d_H(v) \ge (1-\varepsilon)\binom{n-1}{k-1}$.*

We use the following result of Frankl [5, Theorem 10.3].

Theorem 2.3. *For integers $t \ge 1$ and $n \ge 2k$, if an n-vertex k-uniform hypergraph H has more than $t\binom{n-1}{k-1}$ edges, then H has a matching of size $t+1$.*

We also use the following result of Balogh, Bohman and Mubayi [1]. If an intersecting k-uniform hypergraph is a subgraph of a full k-star, then it is called *trivial*, otherwise *non-trivial*. Moreover, we say that a k-uniform hypergraph H is covered by a set $X \subseteq \binom{V(H)}{2}$ of pairs of vertices of H if for every hyperedge e of H, there is a pair $\{x,y\} \in X$ such that $\{x,y\} \subseteq e$.

Lemma 2.4. *([1]) Let H be a non-trivial intersecting k-uniform hypergraph. Then H can be covered by at most k^2-k+1 pairs of vertices.*

3. Proof of Theorem 1.3

Let $k \ge 3$ be an odd integer. Let $\varepsilon := \varepsilon(k) > 0$ be sufficiently small and let $n(k,\varepsilon)$ be a sufficiently large integer. For $n \ge n(k,\varepsilon)$, let H be an n-vertex k-uniform hypergraph with no 2-regular subgraphs. By removing edges if necessary, we may assume that

$$|H| = \binom{n-1}{k-1} + \left\lfloor \frac{n-1}{k} \right\rfloor. \tag{3.1}$$

To prove Theorem 1.3, it is enough to show that H contains a full k-star, because a full k-star with two additional intersecting edges always gives a 2-regular subgraph. To derive a contradiction, we assume that H does not contain any full k-star. Since n is sufficiently large, Theorem 2.2 implies that there is a vertex $v \in V(H)$ such that $d_H(v) \ge \binom{n-1}{k-1} - \varepsilon^3 n^{k-1}$. Let $V' := V(H) \setminus \{v\}$, $H^* := H[V']$ and $\tilde{H} := \{e \setminus \{v\} : |e| = k,\ v \in e \notin H\}$. Note that any $(k-1)$-set $A \subseteq V'$ with $A \notin \tilde{H}$ satisfies that $A \cup \{v\} \in H$. Let

$$x := |\tilde{H}| = \binom{n-1}{k-1} - d_H(v) \le \varepsilon^3 n^{k-1}. \tag{3.2}$$

Since H does not contain a full k-star with center v, we have $x \ge 1$. Then (3.1) and (3.2) imply that

$$|H^*| = x + \left\lfloor \frac{n-1}{k} \right\rfloor. \tag{3.3}$$

An important idea for the proof is that a pair of two intersecting edges in H^* forces $\tilde{H}$ to contain more $(k-1)$-sets (Claim 3.1). Since $x > 0$, there exists a pair of intersecting edges in H^*, Claim 3.1 implies that the value of $x = |\tilde{H}|$ is much larger than 0. However, by (3.3), a larger value of x guarantees more pairs of intersecting edges in H^*, which, in turn, implies that the value of x is even larger. Repeating this argument gives a contradiction as x cannot be too large by (3.2).

To turn this idea into a mathematical proof, we need to prove some technical claims. Here, we give a brief outline of our proof. We start with some simple but useful claims (Section 3.1), and in particular, we show that $x \geq n - 2k + 1$ (Claim 3.3). Combining it with (3.2), we may assume that there exists an integer $2 \leq \ell \leq k-1$ such that

$$\varepsilon^3 n^{\ell-1} \leq x \leq \varepsilon^3 n^{\ell}.$$

We next find pairwise disjoint $(\ell-1)$-sets $S_1, S_2, \ldots, S_{2k} \subseteq V'$ such that $d_{\tilde{H}}(S_i) \leq \binom{k}{\ell-1} x / \binom{n-1}{\ell-1}$ for $i \in [2k]$, which play an important role in the proof. Let $\mathcal{T} := \bigcup_{i=1}^{2k} N_{\tilde{H}}(S_i)$ be a collection of $(k-\ell)$-sets in V'. Let $H_1 := \{e \in H^* : \exists T \in \mathcal{T}$ such that $T \subset e\}$ and let $H_0 := H^* \setminus H_1$. Our goal is to show that $|H_1| \leq \varepsilon x$ (Section 3.2) and $|H_0| < (1-\varepsilon)x + \lfloor \frac{n-1}{k} \rfloor$ (Section 3.3), which together imply that $|H^*| = |H_1| + |H_0| < x + \lfloor \frac{n-1}{k} \rfloor$, contradicting (3.3). In fact, the technical parts are Sections 3.2 and 3.3, in which the argument is essentially a clever double counting also used in [12]. However, as mentioned in Section 1, our case is more complicated than that in [12], so we have to do a more careful analysis (including introducing ℓ and $\mathcal{T}$) and use some new tricks (e.g. analyzing the intersecting property of a certain family and utilizing Lemma 2.4).

3.1. Preparation

First we prove the following easy claim.

Claim 3.1. *Assume that we have $e_1, e_2 \in H^*$, $A \subseteq V'$ and $\{S, S'\}$ such that*

- $A \cap (e_1 \cup e_2) = \emptyset$,
- $|A| = |e_1 \cap e_2| - 1$,
- $\{S, S'\}$ *is an equipartition of* $e_1 \triangle e_2$.

Then either $A \cup S \in \tilde{H}$ or $A \cup S' \in \tilde{H}$.

Proof. If both $(k-1)$-sets $A \cup S$ and $A \cup S'$ are not in $\tilde{H}$, then

$$e_1, e_2, A \cup S \cup \{v\} \text{ and } A \cup S' \cup \{v\}$$

form a 2-regular subgraph of H, a contradiction. □

Now we prove the following two claims regarding lower bounds on x.

Claim 3.2. *Let $t \in [k-1]$. If H^* contains two edges e_1, e_2 such that $|e_1 \cap e_2| = t$, then*

$$x \geq \frac{1}{2}\binom{2k-2t}{k-t}\binom{n-2k+t-1}{t-1}.$$

Proof. Suppose $e_1, e_2 \in H^*$ such that $|e_1 \cap e_2| = t$. Consider a set $A \in \binom{V' \setminus (e_1 \cup e_2)}{t-1}$ and an equipartition $\{S, S'\}$ of $e_1 \triangle e_2$. For each A and $\{S, S'\}$, Claim 3.1 implies that $A \cup S \in \tilde{H}$ or $A \cup S' \in \tilde{H}$. Moreover, distinct choices of $(A, \{S, S'\})$ give us distinct $(k-1)$-sets in $\tilde{H}$.

Since there are $\binom{n-2k+t-1}{t-1}$ distinct choices of A and $\frac{1}{2}\binom{2k-2t}{k-t}$ distinct choices of $\{S, S'\}$, we have $x = |\tilde{H}| \geq \frac{1}{2}\binom{2k-2t}{k-t}\binom{n-2k+t-1}{t-1}$. $\square$

Claim 3.3. *The hypergraph H^* contains two edges e_1, e_2 such that $|e_1 \cap e_2| \geq 2$. Moreover, $x \geq n - 2k + 1$.*

Proof. Assume H^* does not contain such two edges. Then for any $u \in V'$ and $S, S' \in N_{H^*}(u)$, we have $S \cap S' = \emptyset$. If there are two $(k-1)$-sets $S, S' \in N_{H^*}(u)$ such that $S, S' \notin \tilde{H}$, then

$$S \cup \{u\}, S' \cup \{u\}, S \cup \{v\} \text{ and } S' \cup \{v\}$$

form a 2-regular subgraph of H, a contradiction. Thus for any $u \in V'$, we have $|N_{H^*}(u) \cap \tilde{H}| \geq |N_{H^*}(u)| - 1$. Moreover, by our assumption, we have $N_{H^*}(u) \cap N_{H^*}(u') = \emptyset$ for any distinct $u, u' \in V'$. Thus

$$x = |\tilde{H}| \geq \sum_{u \in V'} |N_{H^*}(u) \cap \tilde{H}| \geq \sum_{u \in V'} (d_{H^*}(u) - 1) = k|H^*| - (n-1) \overset{(3.3)}{\geq} kx - (k-1).$$

Since $k \geq 3$, we get $x \leq 1$. However, the assumption that $x \geq 1$ and (3.3) imply that there are two edges $e_1, e_2 \in H^*$ with $|e_1 \cap e_2| \geq 1$. So by Claim 3.2, we have $x \geq \frac{1}{2}\binom{2k-2}{k-1}\binom{n-2k}{0} \geq 3$, a contradiction. Thus H^* contains two edges e_1, e_2 with $|e_1 \cap e_2| \geq 2$. Hence Claim 3.2 implies that $x \geq n - 2k + 1$. $\square$

By (3.2) and Claim 3.3, there exists an integer ℓ such that

$$\varepsilon^3 n^{\ell-1} \leq x \leq \varepsilon^3 n^{\ell} \tag{3.4}$$

and $2 \leq \ell \leq k-1$. Throughout the rest of the paper, ℓ denotes such integer satisfying (3.4).

The following claim finds $2k$ pairwise disjoint $(\ell-1)$-sets which have low degree in $\tilde{H}$.

Claim 3.4. *There are pairwise disjoint $(\ell-1)$-sets $S_1, S_2, \dots, S_{2k} \subseteq V'$ such that $d_{\tilde{H}}(S_i) \le \binom{k}{\ell-1}x/\binom{n-1}{\ell-1}$ for $i \in [2k]$.*

Proof. Let $F := \{S \in \binom{V'}{\ell-1} : d_{\tilde{H}}(S) \le \binom{k}{\ell-1}x/\binom{n-1}{\ell-1}\}$ and $F' := \binom{V'}{\ell-1} \setminus F$. So it suffices to find a matching of size $2k$ in F. Then

$$\binom{k-1}{\ell-1}x = \sum_{S \in \binom{V'}{\ell-1}} d_{\tilde{H}}(S) \ge 0 \cdot |F| + \frac{\binom{k}{\ell-1}x}{\binom{n-1}{\ell-1}}|F'|.$$

So we have

$$|F'| \le \frac{\binom{k-1}{\ell-1}}{\binom{k}{\ell-1}}\binom{n-1}{\ell-1} = \frac{k-\ell+1}{k}\binom{n-1}{\ell-1} \le \frac{k-1}{k}\binom{n-1}{\ell-1},$$

as $\ell \ge 2$. Since $|F| + |F'| = \binom{n-1}{\ell-1}$, we have $|F| \ge \frac{1}{k}\binom{n-1}{\ell-1} > 2k\binom{|V'|-1}{\ell-2}$. Then by Theorem 2.3, F contains a matching $\{S_1, \dots, S_{2k}\}$ of size $2k$ as desired. □

Let $S_1, \dots, S_{2k}$ be a collection of pairwise disjoint $(\ell-1)$-sets as in Claim 3.4. Let $\mathcal{T} := \bigcup_{i=1}^{2k} N_{\tilde{H}}(S_i)$ be a collection of $(k-\ell)$-sets in V'. So we have

$$|\mathcal{T}| \le \sum_{i=1}^{2k} |N_{\tilde{H}}(S_i)| \le \frac{2k\binom{k}{\ell-1}x}{\binom{n-1}{\ell-1}} \le 2k^{k+1}n^{1-\ell}x. \tag{3.5}$$

Note that for any $(k-\ell)$-set $T \notin \mathcal{T}$, we have $T \cup S_i \cup \{v\} \in H$ if $T \cap S_i = \emptyset$. Let $W = \bigcup_{T \in \mathcal{T}} T$, then

$$|W| \le (k-\ell)|\mathcal{T}| \overset{(3.5)}{\le} 2k^{k+2}n^{1-\ell}x \overset{(3.4)}{\le} \varepsilon^2 n. \tag{3.6}$$

Let $H_1 := \{e \in H^* : \exists T \in \mathcal{T} \text{ such that } T \subset e\}$ and let $H_0 := H^* \setminus H_1$.

We finish this subsection with an essential claim that bounds the degrees of vertex sets of size at most ℓ from above.

Claim 3.5. *Any ℓ-set $L \subseteq V'$ satisfies that $|N_{H^*}(L) \setminus \mathcal{T}| \le 1$. Moreover, for any set $B \subseteq V'$ with $|B| = b \le \ell$, it satisfies*

$$d_{H_0}(B) \le \binom{n-1-b}{\ell-b} \text{ and } d_{H^*}(B) \le \binom{n-1-b}{\ell-b}(1+|\mathcal{T}|).$$

Proof. Suppose that there exists an ℓ-set $L \subseteq V'$ such that $|N_{H^*}(L) \setminus \mathcal{T}| \ge 2$. Then there are two distinct $(k-\ell)$-sets $E_1, E_2 \in N_{H^*}(L) \setminus \mathcal{T}$. Since $2 \le \ell \le k-1$, there exists $i \in [2k]$ such that S_i is disjoint from $E_1 \cup E_2 \cup L$. Also we choose a (possibly empty) set

$A \subseteq V' \setminus W$ such that $|A| = |E_1 \cap E_2|$ and $A \cap (E_1 \cup E_2 \cup L \cup S_i) = \emptyset$. This choice is possible since

$$|V' \setminus (W \cup E_1 \cup E_2 \cup L \cup S_i)| \overset{(3.6)}{\geq} (n-1) - \varepsilon^2 n - 2k + \ell - (\ell - 1) \geq k.$$

We claim that both $S_i \cup A \cup (E_1 \setminus E_2)$ and $S_i \cup A \cup (E_2 \setminus E_1)$ are not in $\tilde{H}$. Indeed, if $A = \emptyset$, then $E_j \setminus E_{3-j} = E_j$ for $j \in [2]$. Since $E_j \notin \mathcal{T} = \bigcup_{i=1}^{2k} N_{\tilde{H}}(S_i)$, we obtain $S_i \cup E_j = S_i \cup (E_j \setminus E_{3-j}) \notin \tilde{H}$ for $j \in [2]$. If $A \neq \emptyset$, then since $A \cap W = \emptyset$, we have for $j \in [2]$, $A \cup (E_j \setminus E_{3-j}) \nsubseteq W$. So $A \cup (E_j \setminus E_{3-j}) \notin \mathcal{T}$ and thus $S_i \cup A \cup (E_j \setminus E_{3-j}) \notin \tilde{H}$ for $j \in [2]$. Thus

$$L \cup E_1, L \cup E_2, \{v\} \cup S_i \cup A \cup (E_1 \setminus E_2) \text{ and } \{v\} \cup S_i \cup A \cup (E_2 \setminus E_1)$$

form a 2-regular subgraph of H, a contradiction. Thus the first part of the claim holds.

For any $B \subseteq V'$ with $|B| = b \leq \ell$,

$$d_{H_0}(B) \leq \sum_{B \subseteq L, |L| = \ell} d_{H_0}(L) \leq \sum_{B \subseteq L, |L| = \ell} |N_{H^*}(L) \setminus \mathcal{T}| \leq \sum_{B \subseteq L, |L| = \ell} 1 = \binom{n-1-b}{\ell - b}.$$

Since $d_{H^*}(L) \leq |N_{H^*}(L) \setminus \mathcal{T}| + |\mathcal{T}| \leq 1 + |\mathcal{T}|$ for any ℓ-set L, we also have

$$d_{H^*}(B) \leq \sum_{B \subseteq L, |L| = \ell} d_{H^*}(L) \leq \sum_{B \subseteq L, |L| = \ell} (1 + |\mathcal{T}|) = \binom{n-1-b}{\ell - b}(1 + |\mathcal{T}|). \quad \square$$

In the next two subsections, we show that $|H_1| \leq \varepsilon x$ (Section 3.2) and $|H_0| < (1-\varepsilon)x + \lfloor \frac{n-1}{k} \rfloor$ (Section 3.3), which together imply that $|H^*| = |H_1| + |H_0| < x + \lfloor \frac{n-1}{k} \rfloor$. This contradicts (3.3) and thus completes the proof of Theorem 1.3.

3.2. Size of H_1

In this subsection, we show that $|H_1| \leq \varepsilon x$. We first consider the case $\ell \leq k - 2$.

Claim 3.6. *If $\ell \leq k - 2$, then $|H_1| \leq \varepsilon x$.*

Proof. We first claim that we may assume that $|\mathcal{T}| > 0$, $|H_1| \geq 3|\mathcal{T}|$ and $\ell \geq (k+1)/2$. Indeed, since $|\mathcal{T}| = 0$ implies $|H_1| = 0 \leq \varepsilon x$, we may assume that $|\mathcal{T}| > 0$. If $|H_1| < 3|\mathcal{T}|$, then by (3.5), $|H_1| \leq 6k^{k+1} n^{1-\ell} x \leq \varepsilon x$ because $\ell \geq 2$ and n is sufficiently large. Thus we may assume that $|H_1| \geq 3|\mathcal{T}|$. Finally, since $|H_1| \geq 3|\mathcal{T}| > |\mathcal{T}|$, there is a $(k-\ell)$-set $T \in \mathcal{T}$ which is a subset of two distinct edges e_1, e_2 of H_1. Since $|e_1 \cap e_2| \geq |T| \geq k - \ell$, Claim 3.2 and (3.4) implies that

$$\frac{1}{2}\binom{2\ell}{\ell}\binom{n-k-\ell-1}{k-\ell-1} \leq x \overset{(3.4)}{\leq} \varepsilon^3 n^\ell.$$

Since n is sufficiently large and ε is small, this implies that $\ell > k - \ell - 1$. Thus we have $\ell \geq (k+1)/2$ since k is odd.

Let p be the number of tuples $(T, \{e_1, e_2\}, f)$ with the following properties.

(P.1.1) $T \in \mathcal{T}$, $\{e_1, e_2\} \in \binom{H_1}{2}$ and $f \in \tilde{H}$,
(P.1.2) $T \subseteq e_1 \cap e_2$,
(P.1.3) $f \cap (e_1 \cap e_2) = \emptyset$ and $\{|f \cap e_1|, |f \cap e_2|\} = \{1, |e_2 \setminus e_1| - 1\}$.

First we find a lower bound on p. Fix a $(k-\ell)$-set T in $\mathcal{T}$ and a pair $\{e_1, e_2\} \in P(T)$, where $P(T) := \{\{e_1, e_2\} \in \binom{H_1}{2} : T \subseteq e_1 \cap e_2\}$. Let A be an arbitrary set of size $|e_1 \cap e_2| - 1$ in $V' \setminus (e_1 \cup e_2)$ and let $\{S, S'\}$ be an equipartition of $e_1 \triangle e_2$ such that $|S \cap e_1| = 1$. Then Claim 3.1 implies that one of $A \cup S$ and $A \cup S'$ belongs to $\tilde{H}$ and it satisfies (P.1.3). Note that distinct choices of $(A, \{S, S'\})$ give us distinct $(k-1)$-sets in $\tilde{H}$.

We have $|e_1 \cap e_2| \geq |T| = k - \ell$. Since there are at least $\binom{n-2k}{|e_1 \cap e_2| - 1} \geq \binom{n-2k}{k-\ell-1}$ distinct choices of A and at least one choice of equipartition $\{S, S'\}$ with $|S \cap e_1| = 1$, we obtain

$$p \geq \sum_{T \in \mathcal{T}} \sum_{\{e_1, e_2\} \in P(T)} \binom{n-2k}{k-\ell-1} = \binom{n-2k}{k-\ell-1} \sum_{T \in \mathcal{T}} \binom{d_{H_1}(T)}{2}$$
$$\geq \binom{n-2k}{k-\ell-1} |\mathcal{T}| \binom{\frac{1}{|\mathcal{T}|} \sum_{T \in \mathcal{T}} d_{H_1}(T)}{2} \geq \binom{n-2k}{k-\ell-1} |\mathcal{T}| \binom{|H_1|/|\mathcal{T}|}{2}.$$

Here, we get the penultimate inequality from the convexity of the real function $f(z) = \binom{z}{2} = z(z-1)/2$. Since $|H_1| \geq 3|\mathcal{T}|$, we have that $\binom{|H_1|/|\mathcal{T}|}{2} \geq |H_1|^2/(3|\mathcal{T}|^2)$ and thus

$$p \geq \frac{1}{3} \binom{n-2k}{k-\ell-1} \frac{|H_1|^2}{|\mathcal{T}|}. \tag{3.7}$$

Now we find an upper bound of p. Clearly there are at most $x = |\tilde{H}|$ choices for the $(k-1)$-set f and there are at most $|\mathcal{T}|$ choices for $T \in \mathcal{T}$. For given f, we choose two disjoint subsets $S_1, S_2 \subseteq f$ with $|S_1| = 1$. There are at most $(k-1)2^{k-2}$ ways to choose S_1 and S_2.

Assume that f, T, S_1 and S_2 are fixed, and we count the number of pairs of distinct edges $e_1, e_2 \in H_1$ such that $T \subseteq e_1 \cap e_2$, $e_1 \cap f = S_1, e_2 \cap f = S_2$, and $|e_2 \setminus e_1| - 1 = |e_2 \cap f| = |S_2|$. We choose $e_1 \in H_1$ with $T \cup S_1 \subseteq e_1$, and a set $B \subseteq e_1 \setminus (T \cup S_1)$ with $|B| = k - |T| - |S_2| - 1$. By Claim 3.5, there are $d_{H_1}(T \cup S_1) \leq \binom{n}{2\ell-k-1}(|\mathcal{T}| + 1)$ ways to choose such an edge e_1 and there are at most 2^k ways to choose such a set B. Then we choose $e_2 \in H_1$ such that $T \cup B \cup S_2 \subseteq e_2$ and $e_1 \cap e_2 = T \cup B$. There are at most $d_{H_1}(T \cup B \cup S_2) \leq 1$ way to choose such a set e_2 by Claim 3.5. Thus for fixed f, T, S_1, S_2, the number of choices of e_1, e_2 is at most $2^k \binom{n}{2\ell-k-1}(|\mathcal{T}| + 1)$. Thus we obtain

$$p \le \sum_{f\in\tilde{H}} \sum_{T\in\mathcal{T}} \sum_{S_1,S_2} 2^k \binom{n}{2\ell-k-1}(|\mathcal{T}|+1)$$
$$\le x|\mathcal{T}|(k-1)2^{k-2}\cdot 2^k\binom{n}{2\ell-k-1}(|\mathcal{T}|+1) \le k2^{2k}\binom{n}{2\ell-k-1}|\mathcal{T}|^2 x. \quad (3.8)$$

Note that the third sum is over S_1, S_2 satisfying $|S_1| = 1, S_1 \subseteq f, S_2 \subseteq f \setminus S_1$. From (3.7) and (3.8), we get

$$|H_1|^2 \le 3k2^{2k}\binom{n}{2\ell-k-1}\binom{n-2k}{k-\ell-1}^{-1}|\mathcal{T}|^3 x \le k^{5k}n^{3\ell-2k}|\mathcal{T}|^3 x \overset{(3.5)}{\le} k^{10k}n^{3-2k}x^4.$$

Thus, we get

$$|H_1| \le k^{5k}n^{3/2-k}x^2 \overset{(3.4)}{\le} k^{5k}\varepsilon^3 n^{\ell+3/2-k}x \le \varepsilon x,$$

because $\ell \le k-2$. □

Now assume that $\ell = k-1$. In this case $\mathcal{T}$ is a collection of singletons, any vertex in W belongs to $\mathcal{T}$ and $|W| = |\mathcal{T}|$. We partition $H_1 = G_1 \cup G_2 \cup \cdots \cup G_k$, where $G_i = \{e \in H_1 : |e \cap W| = i\}$ for each $i \in [k]$. Since $\ell = k-1$, the fact that ε is small and (3.5) imply

$$3|W|^{k-1} \le 3(2k^{k+1}n^{2-k}x)^{k-1} \le k^{2k^2}(xn^{1-k})^{k-2}x \overset{(3.4)}{\le} k^{2k^2}(\varepsilon^3)^{k-2}x \le \varepsilon x/k. \quad (3.9)$$

Now we show that $|G_i| \le \varepsilon x/k$ for all $i \in [k]$ which together imply that $|H_1| \le \varepsilon x$.

Claim 3.7. *$|G_k| \le \varepsilon x/k$ and $|G_{k-1}| \le \varepsilon x/k$.*

Proof. First, since G_k does not contain any 2-regular subgraphs, by Theorem 2.1, there exists $n_0 = n_0(k,2)$ such that if $|W| \ge n_0$ then $|G_k| \le 3\binom{|W|}{k-1}$. If $|W| \le n_0$, then (3.4) together with Claim 3.3 implies $|G_k| \le n_0^k < \varepsilon x/k$ since n is large enough. Otherwise $|W| > n_0$, then $|G_k| \le 3\binom{|W|}{k-1} \le \varepsilon x/k$ by (3.9). Second, since $\ell = k-1$ we have

$$|G_{k-1}| = \sum_{L\in\binom{W}{k-1}} |N_{H^*}(L)\setminus\mathcal{T}| \overset{\text{Claim 3.5}}{\le} \sum_{L\in\binom{W}{k-1}} 1 \le |W|^{k-1} \overset{(3.9)}{\le} \varepsilon x/k. \quad \square$$

Now we estimate $|G_i|$ for $1 \le i \le k-2$.

Claim 3.8. *$|G_i| \le \varepsilon x/k$ for $i \in [k-2]$.*

Proof. Assume $|G_i| > \varepsilon x/k$ for some $i \in [k-2]$, then (3.9) implies that $|G_i| \ge 3|W|^{k-1}$. Let p_i be the number of the tuples $(S, \{e_1, e_2\}, f)$ with the following properties.

(P.2.1) $e_1, e_2 \in G_i$ and $f \in \tilde{H}$,
(P.2.2) $S \in \binom{W}{i}$ and $S \subseteq e_1 \cap e_2$,
(P.2.3) $f \cap (e_1 \cap e_2) = \emptyset$, and $\{|f \cap e_1|, |f \cap e_2|\} = \{1, |e_1 \setminus e_2| - 1\}$.

Let $P_i(S) := \{\{e_1, e_2\} \in \binom{G_i}{2} : S \subseteq e_1 \cap e_2\}$. By convexity, we have

$$\sum_{S \in \binom{W}{i}} \sum_{\{e_1,e_2\} \in P_i(S)} 1 \geq \sum_{S \in \binom{W}{i}} \binom{d_{G_i}(S)}{2} \geq \binom{|W|}{i} \binom{|G_i|/\binom{|W|}{i}}{2} \geq \frac{|G_i|^2}{3\binom{|W|}{i}}, \quad (3.10)$$

where we used $\sum_{S \in \binom{W}{i}} d_{G_i}(S) = |G_i|$ and $|G_i| \geq 3|W|^{k-1} \geq 3\binom{W}{i}$.

Consider a set $S \subseteq W$ of size i, and a pair $\{e_1, e_2\} \in P_i(S)$. Let A be an arbitrary set of size $|e_1 \cap e_2| - 1$ in $V' \setminus (e_1 \cup e_2)$, and let A_1, A_2 be a partition of $e_1 \triangle e_2$ such that $|A_1| = 1$. The number of ways to choose A is at least $\binom{n-2k}{|e_1 \cap e_2|-1} \geq \binom{n-2k}{i-1}$ and the number of ways to choose A_1, A_2 is at least one. By Claim 3.1, at least one of $A \cup A_1 \in \tilde{H}$ and $A \cup A_2 \in \tilde{H}$ holds. Then either $(S, \{e_1, e_2\}, A \cup A_1)$ or $(S, \{e_1, e_2\}, A \cup A_2)$ satisfies (P.2.1)–(P.2.3). Since distinct choices of $(S, \{e_1, e_2\}, A, \{A_1, A_2\})$ give us distinct tuples, we have

$$p_i \geq \sum_{S \in \binom{W}{i}} \sum_{\{e_1,e_2\} \in P_i(S)} \sum_{A} \sum_{A_1, A_2} 1 \geq \sum_{S \in \binom{W}{i}} \sum_{\{e_1,e_2\} \in P_i(S)} \binom{n-2k}{i-1} \overset{(3.10)}{\geq} \binom{n-2k}{i-1} \frac{|G_i|^2}{3\binom{|W|}{i}}. \quad (3.11)$$

Now we find an upper bound of p_i. Clearly there are at most $x = |\tilde{H}|$ choices of $f \in \tilde{H}$ and at most $\binom{|W|}{i}$ choices of $S \in \binom{W}{i}$ with $S \cap f = \emptyset$. For given f and S, we choose two disjoint sets $A_1, A_2 \subseteq f$ with $|A_1| = 1$. There are at most $(k-1)2^{k-2}$ ways to choose such A_1 and A_2.

Assume f, S, A_1, A_2 are given, and we count the number of pairs $\{e_1, e_2\} \in P_i(S)$ such that $e_1 \cap f = A_1$, $e_2 \cap f = A_2$ and $|e_2 \setminus e_1| - 1 = |e_2 \cap f| = |A_2|$. We choose $e_1 \in G_i$ such that $S \cup A_1 \subseteq e_1$ and $e_1 \setminus (S \cup A_1) \subseteq V' \setminus W$, and the number of ways to choose such e_1 is at most

$$\sum_{S \cup A_1 \subseteq L \in \binom{V'}{k-1}} |N_{H^*}(L) \setminus W| \overset{\text{Claim 3.5}}{\leq} \sum_{S \cup A_1 \subseteq L \in \binom{V'}{k-1}} 1 = \binom{n-i-2}{k-i-2}.$$

We also choose a set $B \subseteq e_1 \setminus (S \cup A_1)$ with $|B| = k - |S| - |A_2| - 1$. There are at most 2^k ways to choose such a set B. Then we choose $e_2 \in G_i$ such that $S \cup B \cup A_2 \subseteq e_2$, $e_1 \cap e_2 = S \cup B$ and $e_2 \setminus (S \cup B \cup A_2) \subseteq V' \setminus W$, and the number of ways to choose such e_2 is at most $|N_{H^*}(S \cup B \cup A_2) \setminus W| \leq 1$ by Claim 3.5. Overall, for fixed f, S, A_1, A_2, the number of choices of e_1, e_2 is at most $2^k \binom{n-i-2}{k-i-2}$. Thus we obtain

$$\begin{aligned} p_i &\le \sum_{f\in\tilde{H}} \sum_{S\in\binom{W}{i}} \sum_{A_1,A_2} 2^k \binom{n}{k-i-2} \\ &\le x\binom{|W|}{i}(k-1)2^{k-2}\cdot 2^k\binom{n}{k-i-2} \le k2^{2k}x\binom{|W|}{i}\binom{n}{k-i-2}. \end{aligned} \tag{3.12}$$

Note that the third sum is over A_1, A_2 satisfying $|A_1| = 1, A_1 \subseteq f, A_2 \subseteq f \setminus A_1$.

From (3.11) and (3.12) and the fact that $|W| = |\mathcal{T}|$, we obtain

$$\begin{aligned} |G_i|^2 &\le 3k2^{2k}\binom{|W|}{i}^2\binom{n}{k-i-2}\binom{n-2k}{i-1}^{-1}x \\ &\overset{(3.5)}{\le} k^{3k}(2k^{k+1}n^{2-k}x)^{2i}n^{k-2i-1}x \\ &\le k^{10k^2}n^{-(2i-1)(k-1)}x^{2i+1} \\ &\overset{(3.4)}{\le} k^{10k^2}n^{-(2i-1)(k-1)}(\varepsilon^3 n^{k-1})^{2i-1}x^2 \le k^{10k^2}\varepsilon^3x^2 < \varepsilon^2x^2/k^2. \end{aligned}$$

This contradicts that $|G_i| > \varepsilon x/k$. Thus the claim holds. □

3.3. Size of H_0

At last we show that $|H_0| < (1-\varepsilon)x + \lfloor \frac{n-1}{k} \rfloor$. Assume to the contrary, that

$$|H_0| \ge (1-\varepsilon)x + \left\lfloor \frac{n-1}{k} \right\rfloor. \tag{3.13}$$

For any $u \in V'$, let $F_u := N_{H_0}(u) \setminus \tilde{H}$. We first observe that F_u is an intersecting family.

Claim 3.9. *For any $u \in V'$, F_u forms an intersecting family.*

Proof. If not, then there are two disjoint $(k-1)$-sets $A, A' \in F_u = N_{H_0}(u) \setminus \tilde{H}$. Since $A, A' \notin \tilde{H}$,

$$A \cup \{u\}, A' \cup \{u\}, A \cup \{v\} \text{ and } A' \cup \{v\}$$

form a 2-regular subgraph of H, a contradiction. □

If $f \in \tilde{H}$ belongs to $N_{H_0}(u) \cap N_{H_0}(u')$ for distinct $u, u' \in V'$, then fix any $L \in \binom{f}{\ell}$, we have $d_{H_0}(L) \ge 2$, contradicting Claim 3.5. Thus $f \in \tilde{H}$ belongs to $N_{H_0}(u)$ for at most one $u \in V'$, which implies that

$$x = |\tilde{H}| \ge \sum_{u\in V'} |N_{H_0}(u) \cap \tilde{H}| = \sum_{u\in V'} (d_{H_0}(u) - |F_u|). \tag{3.14}$$

Note that this implies that $\ell \geq 3$ and $k \geq 5$. In fact, if $\ell = 2$, then Claim 3.5 implies that $d_{H_0}(\{u, u'\}) \leq 1$ for any two distinct vertices $u, u' \in V'$, i.e., any two edges in H_0 share at most one vertex. Thus $N_{H_0}(u)$ forms a matching. By Claim 3.9, $|F_u| \leq 1$ as F_u is an intersecting subfamily of a matching. By (3.14),

$$x \geq \sum_{u \in V'} (d_{H_0}(u) - 1) = k|H_0| - (n-1) \overset{(3.13)}{\geq} k(1-\varepsilon)x - (k-1).$$

Thus $x \leq 1$, contradicting (3.4). Thus $\ell \geq 3$. Since $3 \leq \ell \leq k-1$ by (3.4) and k is odd, we have $k \geq 5$.

Let

$$X := \{u \in V' : F_u \text{ is a trivial intersecting family}\}$$

and for $u \in X$, let $p(u)$ be a vertex in V' such that every $(k-1)$-set in F_u contains $p(u)$. We claim that

$$\sum_{u \in X} |F_u| \geq (1-\varepsilon)(k-1)|H_0|. \tag{3.15}$$

We first show that for $u \notin X$, $|F_u| \leq k^2 n^{\ell-3}$. Indeed, since $u \notin X$, F_u is a non-trivial intersecting $(k-1)$-uniform family. By Lemma 2.4, there are pairs of vertices $w_1 w_1', \ldots, w_t w_t'$ with $t \leq (k-1)^2 - (k-1) + 1 \leq k^2$ which together cover all $(k-1)$-sets in F_u. Since $\ell \geq 3$, Claim 3.5 implies

$$|F_u| \leq \sum_{i=1}^{t} d_{H_0}(\{u, w_i, w_i'\}) \leq k^2 \binom{n-4}{\ell-3} \leq k^2 n^{\ell-3}.$$

Then note that $\sum_{u \in V'} d_{H_0}(u) = k|H_0|$. From (3.14), we get

$$\begin{aligned} x &\geq \sum_{u \in V'} (d_{H_0}(u) - |F_u|) \geq \sum_{u \in V'} d_{H_0}(u) - \sum_{u \in X} |F_u| - \sum_{u \in V' \setminus X} |F_u| \\ &\geq k|H_0| - \sum_{u \in X} |F_u| - \sum_{u \in V' \setminus X} k^2 n^{\ell-3} \geq k|H_0| - \sum_{u \in X} |F_u| - k^2 n^{\ell-2}. \end{aligned}$$

Since n is sufficiently large, (3.4) implies that $k^2 n^{\ell-2} \leq \varepsilon^4 n^{\ell-1} \leq \varepsilon x$. Thus we get

$$\sum_{u \in X} |F_u| \geq k|H_0| - x - \varepsilon x \overset{(3.13)}{\geq} k|H_0| - \frac{(1+\varepsilon)|H_0|}{1-\varepsilon} \geq (1-\varepsilon)(k-1)|H_0|$$

as $k \geq 5$. So (3.15) is proved.

For $t \in [k-1]$, let q_t be the number of the tuples $(u, \{e_1, e_2\}, f)$ with the following properties.

(Q1)$_t$ $u \in X$, $e_i \setminus \{u\} \in F_u$ for $i \in [2]$ and $|e_1 \cap e_2| = t$,
(Q2) $f \in \tilde{H}$, $f \cap (e_1 \cap e_2) = \emptyset$,
(Q3) $\{|f \cap e_1|, |f \cap e_2|\} = \{1, |e_1 \setminus e_2| - 1\}$.

For $u \in X$ and $t \in [k-1]$, we let

$$F_u^t := \{\{e_1, e_2\} : e_i \setminus \{u\} \in F_u \text{ for } i \in [2], |e_1 \cap e_2| = t\},$$

and

$$P^t := \{(u, \{e_1, e_2\}) : u \in X, \{e_1, e_2\} \in F_u^t\}.$$

Note that $F_u^1 = \emptyset$ for any $u \in X$ since F_u is an intersecting family. Since $u \in X$, we have $\{u, p(u)\} \subseteq e_1 \cap e_2$ for $(u, \{e_1, e_2\}) \in P^t$. By convexity of function $f(z) = \binom{z}{2} = \frac{z(z-1)}{2}$, we have

$$\begin{aligned}\sum_{t=2}^{k-1} |P^t| &= \sum_{u \in X} \binom{|F_u|}{2} \geq |X| \binom{\frac{1}{|X|}\sum_{u \in X} |F_u|}{2} \\ &\overset{(3.15)}{\geq} \frac{(1-\varepsilon)^2 (k-1)^2 |H_0|^2}{2n} - \frac{1}{2}(1-\varepsilon)(k-1)|H_0| \geq \frac{(1-2\varepsilon)(k-1)^2|H_0|^2}{2n}.\end{aligned} \tag{3.16}$$

Here, we get the last inequality since (3.4) implies $\varepsilon^2 |H_0| \geq \varepsilon^2 x \geq \varepsilon^5 n^2 > 2n$ since $\ell \geq 3$ and n is large.

Now we find a lower bound of q_t. Note that $q_1 = 0$ since F_u is an intersecting family for any $u \in X$. For $2 \leq t \leq k-1$, first fix a vertex $u \in X$ and let $\{e_1, e_2\} \in F_u^t$. We choose a set $A \subseteq V' \setminus (e_1 \cup e_2)$ of size $t-1$. We also choose an equipartition $\{S, S'\}$ of $e_1 \triangle e_2$ such that $|S \cap e_1| = 1$. The number of choices of such $\{S, S'\}$ is $(k-t)^2$. Then Claim 3.1 implies that either $A \cup S$ or $A \cup S'$ belongs to $\tilde{H}$ and it satisfies (Q3) as it plays the role of f. Note that for distinct choices of $(A, \{S, S'\})$, we get distinct $(k-1)$-sets f in $\tilde{H}$.

So for $2 \leq t \leq k-1$,

$$q_t \geq \sum_{u \in X} \sum_{\{e_1, e_2\} \in F_u^t} (k-t)^2 \binom{n-2k}{t-1} = (k-t)^2 \binom{n-2k}{t-1} |P^t|.$$

Since $k \geq 5$ and n is large, $(k-t)^2 \binom{n-2k}{t-1} \geq n(n-2k)$ for $t \geq 3$. Thus we obtain

$$q_2 \geq (k-2)^2 (n-2k)|P^2| \quad \text{and} \quad q_t \geq n(n-2k)|P^t| \ \text{ for } \ 3 \leq t \leq k-1. \tag{3.17}$$

Next we find an upper bound of q_t. Clearly there are at most $x = |\tilde{H}|$ choices for $f \in \tilde{H}$. We choose two disjoint sets $A_1, A_2 \subseteq f$ such that $|A_1| = 1, |A_2| = k-t-1$. The number of

such choices is at most $(k-1)\binom{k-2}{k-t-1}$. Now we choose e_1 in H_0 containing A_1. The number of choices for e_1 is at most $|H_0|$. Once e_1 is chosen, we choose $u \in X \cap (e_1 \setminus A_1)$ such that $\{u, p(u)\} \subseteq e_1$. There are at most $k-1$ such choices for u. Now we choose a $(t-2)$-subset $B \subseteq e_1 \setminus (A_1 \cup \{u, p(u)\})$, and there are $\binom{k-3}{t-2}$ ways to choose such B. For given A_2, B, u, we choose e_2 such that $A_2 \cup B \cup \{u, p(u)\} \subseteq e_2$. Since $|A_2 \cup B \cup \{u, p(u)\}| = k-1$, it contains a subset L of size ℓ, thus Claim 3.5 implies that $d_{H_0}(A_2 \cup B \cup \{u, p(u)\}) \leq d_{H_0}(L) \leq 1$. Thus the number of choices of e_2 is at most 1. Thus we get

$$q_t \leq x(k-1)\binom{k-2}{k-t-1}|H_0|(k-1)\binom{k-3}{t-2}. \tag{3.18}$$

Thus we obtain

$$\begin{aligned}
&(k-2)^2 n(n-2k)\sum_{t=2}^{k-1}|P^t| \\
&\overset{(3.17)}{\leq} nq_2 + (k-2)^2\sum_{t=3}^{k-1} q_t \\
&\overset{(3.18)}{\leq} (k-1)^2(k-2)xn|H_0| + \sum_{t=3}^{k-1}(k-1)^4\binom{k-2}{k-t-1}\binom{k-3}{t-2}x|H_0| \\
&\leq (1+\varepsilon)(k-1)^2(k-2)xn|H_0|.
\end{aligned}$$

Note that we get the last inequality since n is sufficiently large. By (3.16), we get

$$\begin{aligned}
\frac{1}{2}(1-2\varepsilon)(k-2)^2(k-1)^2|H_0|^2(n-2k) &\leq (k-2)^2 n(n-2k)\sum_{t=2}^{k-1}|P^t| \\
&\leq (1+\varepsilon)(k-1)^2(k-2)xn|H_0|.
\end{aligned}$$

Since $|H_0| > 0$, dividing both sides by $(k-2)(k-1)^2|H_0|/2$ gives

$$(1-2\varepsilon)(k-2)|H_0|(n-2k) \leq 2(1+\varepsilon)xn.$$

Since we have $(1-\varepsilon)x \leq |H_0|$ from (3.13),

$$(1-2\varepsilon)(k-2)(1-\varepsilon)x(n-2k) \leq 2(1+\varepsilon)xn.$$

Since $x \geq 1$, we get $(1-3\varepsilon)(k-2)(n-2k) \leq 2(1+\varepsilon)n$, which is a contradiction since $k \geq 5$, ε is small and n is large enough. So (3.13) does not hold and we are done.

4. Concluding remarks

In our proof of Theorem 1.3, except the use of Theorems 2.1 and 2.2, we only use the assumption that H does not contain any 2-regular subgraphs on $2k$ vertices. This motivates the following conjecture.

Conjecture 4.1. *For every integer $k \geq 3$, there exists n_k such that the following holds for all $n \geq n_k$. If H is an n-vertex k-uniform hypergraph with no 2-regular subgraphs on $2k$ vertices, then*

$$|H| \leq \binom{n-1}{k-1} + \left\lfloor \frac{n-1}{k} \right\rfloor.$$

Moreover, equality holds if and only if H is a full k-star with center v together with a maximal matching omitting v.

For $k \geq 4$, Conjecture 1.1 implies Conjecture 4.1. Note that Conjecture 4.1 stands between Conjecture 1.1 and the result on forbidding 2-regular subgraphs. In some sense it is closer to Conjecture 1.1 – because only finitely many (independent of n) configurations are forbidden (in contrast, by forbidding all 2-regular subgraphs, the number of instances forbidden is related to n). By our proof, to show Conjecture 4.1 for odd integers k, it suffices to prove an asymptotical result and a stability result.

In this paper we focused on forbidding 2-regular subgraphs. It is natural to consider hypergraphs without r-regular subgraphs for $r \geq 3$ (see Question 6.9 in [9]). We remark that Construction 6.8 in [9] gives a lower bound on the maximum number of edges in such a hypergraph.

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